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RockPaperScissors.cpp
Assignment 2 (Fix the game!)

Purpose

The original program let the user run a simple session of traditional game "Rock, paper scissors", where they can type down their choices and a bot will display theirs. Although it fulfilled its intended purpose, the program still needed corrections. Therefore, the purpose of this new program is to improve various aspects of its predecessor, such as a better logic flow, upgraded user interface, improved interactions with the user, customization, etc.

Inputs

The program receives the following inputs from the user:

- Username
- Bot selection number
- Player moves for each round
- Whether the user wants to play again

Outputs

The program displays the following outputs:

- A welcome message
- Game rules
- A username prompt
- Bot selection menu
- Warning message for invalid inputs
- A prompt for the player's move
- Bot selected move
- Score updates
- Match results (win, lose or tie)
- Bot dialogues for victory or defeat
- Ask if the user wants to play again
- Goodbye and thanks message

Variables

Variable	Type	Purpose
running	bool	Controls whether the game keeps running.
player_name	string	Saves the username entered by the player.
bot_name	string	Keeps the name of the selected bot.
bot_catalogue	string array	Stores the names for the bot.
chosen_bot	int	Stores the number selected by the user for the bot opponent.
moves	string array	Holds the valid move names: Rock, Paper, and Scissors.
player_move	string	Stores the move entered by the player.
bot_move	int	Stores the random array index used for the bot's move.
bot_pick	string	Saves the bot's chosen move.
player_score	int	Stores the player's score during a match.
bot_score	int	Stores the bot's score during a match.
bot_wins_lines	string array	Holds the dialogue printed when the bot wins a match.
bot_loses_lines	string array	Keeps dialogue printed when the bot loses a match.
answer	string	Saves the user's answer about whether to play again.

Key Design Choices

srand(time(0)): In the early trials of the program, although the bot move generation was random, it was creating a pseudo-pattern that could be tracked down by users who spend long times playing. Therefore, with this function, from the “<ctime>” library, a seed is located in the pattern randomly generated, and, since the seed is the current time, it will never start in the same position twice.

while(running) and game restart: The program uses a while-loop so the user can play multiple matches without restarting the program. In addition, the program resets variables "player_score" and "bot_score" at the beginning of each new match so previous match scores do not affect the next match.

Array optimization: The program uses multiple arrays for bot names, valid moves, and bot dialogues to avoid storing each option in separate variables. In addition, the arrays also improve the logic flow of the code itself while also saving a fair amount of time and lines. For instance, the original version of the code relied on an if-else statement sequence to display the bot choices, whereas this modified version utilizes the array "moves" along with the "rand() % 3" operation to pick a random move from the array and then simply display it in a matter of 3 lines of code.

std::getline() instead of std::cin>>: The program uses "std::getline" for most variable storage lines instead of the common "std::cin>>" because "std::cin" only stores the first word that it reads, which creates space for potential invalid inputs of user entering a correct word followed by a sentence full of gibberish. Instead, "std::getline" can read the whole line entered by the user, improving greatly user input reading, so that input such as "Scissors and pepper" can be read as one full line and rejected as invalid.

<cctype> and case-insensitivity: The program converts the lines in "player_move" to lowercase so inputs such as "ROCK", "Rock", "rock" or "rOCk" can be treated the same way. This is possible due the function "tolower()" from the <cctype> library. Once all characters are turned into lowercase, an if-else statement compares user input to the content in the array "moves" to make it consistent with the bot choices. Similarly, it also converts replay answers to lowercase so inputs such as "YES", "Yes", "y", "N", and "no" can be handled consistently.

nested for-loops: Is worth mentioning that, in order to make the code case-insensitive, a nested for-loop is used. The inner for-loop reads the user line character by character, turning them into lowercase. Moreover, if user input is invalid, not only is a warning message displayed, but also the counter of the outer loop, "rounds" is decreased by 1. This was done to avoid the rounds to ever reach 2 and provoke an accidental win through user's writing errors, forcing the rounds to remain static until the user enters a valid move. Afterall, even if the rounds decrease in the inner loop, they are still increasing by the outer-loop headliner.

"Username and AI character" Addition: In order to make the code more user-friendly while also providing customization and a more interactive experience, the user is now capable of entering a nickname of their choice, being warned of the dangers of providing their real name in online spaces. Also, once their username is choose, the user can also pick a name for the bot, so instead of just playing against the terminal that will give no replies, the user can name the bot after famous AI models from both real world (ChatGPT, Claude and Gemini) and fictional AI

(Skynet, R2D2 and Allied Mastercomputer (AM)). Not only that, but also, once the bot is given a name, it is also provided with dialogues to display when they win or lose a match against the player, so that a sense of competition can be emulated to some degree. The roots of the idea are from the chess.com app that allows you to play against special bots with custom dialogues. Finally, Allied Mastercomputer (AM) is hidden as an easter-egg (like a videogame rewarding user for digging into the material intensely) and can perform a unique exit from the code, given its wrathful personality. In other words, Allied Mastercomputer (AM) is meant to be a bad loser, eager to quit and ruin the experience of the game for everyone.

Program Steps (Algorithm)

1. Display the welcome message and rules.

2. Ask the user to enter a username.

3. Display the bot selection menu and let the user select their opponent based on the number. If the number entered is out of the scope, a warning message alerts the user.

4. Display the selected bot name or the secret bot message.

5. Start the game with 3 rounds.

6. If either player or bot scores at least 2 points, end the round loop early.

7. Ask the user to enter Rock, Paper, or Scissors.

8. If the move is invalid, the program displays a warning and restarts the round until user enters a valid move.

9. Display the bot's chosen move.

10. Compare player and bot choices. If both are the same, no point is given to either side. Otherwise, a point is given to the winner, and the updated score is displayed.

11. After three rounds, the game compares both scores and declares a winner.

12. If the player wins, displays winning message along with their final score and the bot losing dialogue. In the other hand, if the bot wins, the final score is shown along with their winning dialogue.

13. Special addition: If Allied Mastercomputer (AM) loses, a special ending is displayed, and AM will forcefully stop the program being enraged.

14. If the scores are equal, display the tie message.

15. Ask whether the user wants to play again.

16. If the user says yes, the game restarts. Else, the game stops and display good-bye message. Also, if the answer is invalid, the program asks again until a valid answer is given.

Functions

Function: int main()

Purpose: Runs the full “Rock, paper, scissors” game. The spinal cord of the code that holds variables, loop, program output and user input.

Sample Input/Output

Output	Input
||---Let's play 'Rock. Paper. Scissors!!'---|| ||-----------RULES: Write your choice. Your opponent will choose theirs. Best of three rounds wins!!---------------------|| ||Note: If no player has reached 2 points by the end of Round 3, the winner is determined by the highest score.||	
Enter your username (Tip: Please don't use your real name for this type of thing, it's risky):	Buzio
Which artificial intelligence would you like to play with? Select its number to pick it: 0-??? 1-ChatGPT 2-Claude 3-Gemini 4-Skynet 5-R2D2	5
Great!! You are playing against R2D2!!	
Choose rock, paper or scissors:	Scissors
R2D2 has picked: Paper +1 for Buzio Buzio points: 1 R2D2 points: 0 Choose rock, paper or scissors:	Paper
Choose rock, paper or scissors: Paper R2D2 has picked: Paper Oh. No point for either side, I guess. Buzio points: 1	

R2D2 points: 0 Choose rock, paper or scissors:	Rock
R2D2 has picked: Scissors +1 for Buzio Buzio points: 2 R2D2 points: 0 Buzio wins with a total score of 2 over 0	
R2D2: *angry astromech screaming* (translation censored by protocol droids)	
Do you want to play again? [yes/no or y/n]:	No
Thanks for playing 'Rock. Paper. Scissors!!' with us. Bye-Bye!! :D	

It's worth noting that the exact bot moves and final result may change because the bot move is randomly generated.

Testing

Test Case 1: Perfect case-scenario.

Expected Result:

The program accepts ChatGPT as the selected bot, accepts the valid moves, plays the match, displays the score updates, asks whether to play again, and exits when the user enters no.

```
||-----------------------------------Let's play 'Rock. Paper. Scissors!!'------------------------------------||
||----------RULES: Write your choice. Your opponent will choose theirs. Best of three rounds wins!!----------||
||Note: If no player has reached 2 points by the end of Round 3, the winner is determined by the highest score.||

Enter your username (Tip: Please don't use your real name for this type of thing, it's risky): Chiasaki
Which artificial intelligence would you like to play with? Select its number to pick it:
0-???
1-ChatGPT
2-Claude
3-Gemini
4-Skynet
5-R2D2
1
Great!! You are playing against ChatGPT!!

Choose rock, paper or scissors: Paper
ChatGPT has picked: Scissors

+1 for ChatGPT
Chiasaki points: 0
ChatGPT points: 1

Choose rock, paper or scissors: Rock
ChatGPT has picked: Scissors

+1 for Chiasaki
Chiasaki points: 1
ChatGPT points: 1

Choose rock, paper or scissors: Scissor
ChatGPT has picked: Scissors

Oh. No point for either side, I guess.
Chiasaki points: 1
ChatGPT points: 1

No one has the high ground over the other... It's a tie!
Do you want to play again? [yes/no or y/n]: no
Thanks for playing 'Rock. Paper. Scissors!!' with us. Bye-Bye!! :D
```

Test Case 2: Bot selection out of range
Expected Result:
The program rejects 8 as an invalid bot selection, asks for another number, accepts 2 as Claude, and continues the match.

```
||-----------------------------------Let's play 'Rock. Paper. Scissors!!'------------------------------------||
||----------RULES: Write your choice. Your opponent will choose theirs. Best of three rounds wins!!-----------||
||Note: If no player has reached 2 points by the end of Round 3, the winner is determined by the highest score.||

Enter your username (Tip: Please don't use your real name for this type of thing, it's risky): Anthony
Which artificial intelligence would you like to play with? Select its number to pick it:
0-???
1-ChatGPT
2-Claude
3-Gemini
4-Skynet
5-R2D2

8
Chosen number is not assigned to any AI option. Please pick from 0 to 5: 2
Great!! You are playing against Claude!!

Choose rock, paper or scissors: ROCK
Claude has picked: Rock

Oh. No point for either side, I guess.
Anthony points: 0
Claude points: 0

Choose rock, paper or scissors:
```

Test Case 3: Invalid moves
Expected Result:
The program rejects "Scissors and sea salt" because the full line is not exactly a valid move, then accepts Scissors, Rock, and Paper as valid moves.

```
Choose rock, paper or scissors: Scissor and sea salt
Rock Paper or Scissors are the only moves you can make, mate. Try again.
Choose rock, paper or scissors: ScISSor
Gemini has picked: Paper

+1 for Taj
Taj points: 1
Gemini points: 1
```

Test Case 4: AM easter egg and secret ending
Expected Result:
The program accepts Y as yes and starts a new match. It accepts 0 as the secret Allied Mastercomputer (AM) bot choice. It accepts n as no and exits unless the special AM ending stops the game first.

```
Do you want to play again? [yes/no or y/n]: Y
Which artificial intelligence would you like to play with? Select its number to pick it:
0-???
1-ChatGPT
2-Claude
3-Gemini
4-Skynet
5-R2D2
0
Secret character unlocked: Enters Allied Mastercomputer (AM)...

Choose rock, paper or scissors: Paper
Allied Mastercomputer (AM) has picked: Rock

+1 for Ted
Ted points: 1
Allied Mastercomputer (AM) points: 0

Choose rock, paper or scissors: Scissor
Allied Mastercomputer (AM) has picked: Scissors

Oh. No point for either side, I guess.
Ted points: 1
Allied Mastercomputer (AM) points: 0

Choose rock, paper or scissors: Scissor
Allied Mastercomputer (AM) has picked: Scissors

Oh. No point for either side, I guess.
Ted points: 1
Allied Mastercomputer (AM) points: 0

Ted wins with a total score of 1 over 0
Allied Mastercomputer (AM): You think you've won? You have won nothing. The game is over.
Allied Mastercomputer (AM): You are not the one in control, I AM.
*AM attempts to close the program. Doing too much for a simple game... Oh well, bye!!
Thanks for playing 'Rock. Paper. Scissors!!' with us. Bye-Bye!! :D
```